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ADSORPTION OF HYDROGEN ON Pd(100)

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The energetic, kinetic and structural properties of hydrogen chemisorbed on a Pd(100) surface were studied by means of thermal desorption, work function and LEED measurements. Under the applied conditions no interference with bulk dissolution occurs and dissociative adsorption gives rise to a continuous increase of the work function by up to 0.20 eV. The dipole moment of the adsorbate complex is constant up to $\theta \approx 0.9$ and then increases until saturation at $\theta \approx 1.35$ (at 170 K) is reached. The formation of a second adsorbed state at high coverages manifests itself also by a low-temperature shoulder in the thermal desorption spectra and in the variation of the isosteric heat of adsorption, E_{ad}, with coverage: E_{ad} remains practically constant (24.5 kcal/mole) up to $\theta \approx 0.9$ and then decreases. The sticking coefficient is initially rather high ($s_0 \approx 0.5$) and varies with coverage in a way which can be described by a precursor-state model. The preexponential factor for desorption is about 10^{-2} cm^2 atom^{-1} s^{-1}. Desorption follows second order kinetics only at very low coverages, at high θ it exhibits quasi-first order. This effect is attributed to the existence of lateral interactions between adsorbed hydrogen atoms which manifest themselves also in the appearance of a c(2 × 2) LEED pattern at low temperatures. The "extra" diffraction spots attain their maximum intensity at $\theta = 0.5$, and a structural model is proposed whereafter in this phase the H atoms occupy next-nearest neighboring adsorption sites with local fourfold symmetry. Order–disorder transitions were followed by recording the intensity of the half-order spots as a function of temperature at various coverages. The resulting phase diagram exhibits a critical temperature $T_c = 260$ K at $\theta = 0.5$ and is slightly asymmetric with respect to this coverage. The data are analysed in terms of a lattice gas model and estimates for the pairwise interaction energies yield repulsion between nearest neighbors ($w_1 = 0.5$ kcal/mole) and attraction between next-nearest neighbors ($w_2 = -0.3$ kcal/mole). The additional operation of non-pair-wise interactions is made responsible for the asymmetric shape of the phase diagram. Whereas the adsorbed layer is obviously localized at $T \leqslant 270$ K, a detailed analysis of the adsorption entropy reveals that for $T \geqslant 370$ K a rather good description can be obtained with a model of delocalized two-dimensional translation.

1. Introduction

Numerous investigations in the past concentrated on the unique bulk properties of the hydrogen/palladium system [1]. Since Pd is an active catalyst for hydrogena-

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tion reactions also a series of adsorption studies with well-defined clean surfaces has been performed during recent years [2–10]. Frequently the separation of bulk and surface phenomena offered a serious problem. In a recent molecular beam study with Pd(111) Engel and Kuipers [6] demonstrated how the kinetics of these processes can be individually studied. A previous study from our laboratory [2] was concerned with hydrogen adsorption on Pd(111) and (110) single crystal planes in which it was shown how under low pressure conditions thermal desorption and work function ($\Delta\phi$) data can be used for characterization of the adsorption process alone. The present work represents an extension of this approach to the Pd(100) surface where – particularly due to the occurrence of a simple LEED pattern – interesting new features could be recognized. The only other attempt to investigate the H/Pd(100) system was performed several years ago by Szymerska [10] which, however, suffered by insufficient conditions of surface cleanliness.

2. Experimental

The experiments were performed in a UHV system equipped with facilities for LEED (including a Faraday cup for intensity measurements), Auger spectroscopy, work function measurements by the Kelvin technique, and a quadrupole mass spectrometer. Details of the surface cleaning procedure etc. are described in another paper dealing with the adsorption of CO [11].

3. Results

Fig. 1 shows a series of thermal desorption spectra which were recorded with a heating rate of 14 K/s after the sample had been exposed to various doses of H_2 at 170 K. Like in a previous study with the H/Pd(111) and (110) systems [2], the H_2 molecules desorbing below 500 K were initially adsorbed on the surface rather than dissolved in subsurface regions. Since the adsorption energy exceeds the heat of dissolution considerably exposure to gaseous H_2 will at first cause saturation of the adsorbed layer before appreciable amounts penetrate into the bulk [1,2]. Exposure to high amounts of hydrogen causes as in the previous work the appearance of a very broad second desorption maximum centered above 600 K which is attributed to hydrogen atoms back-diffusing from the bulk to the surface where they recombine and desorb.

After desorption of the surface species the work function of the sample returned to the value of the clean surface so that the work function changes (see below) under the present conditions are exclusively caused by *adsorbed* hydrogen. The data of fig. 1 were recorded only for temperatures below 550 K in order to suppress bulk effects.

The area below a thermal desorption trace is proportional to the respective

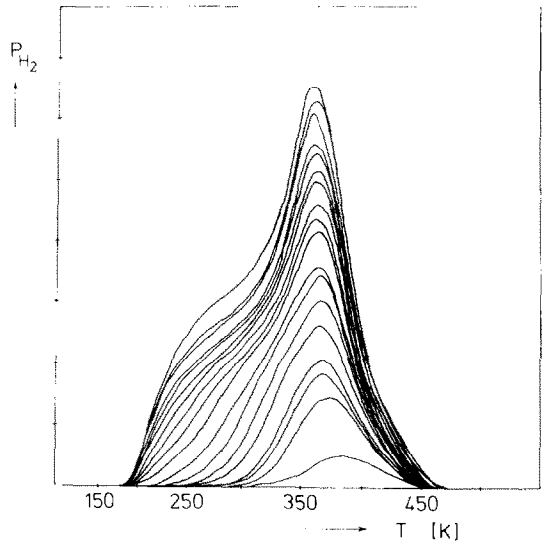


Fig. 1. Series of thermal desorption spectra recorded with a heating rate of 14 K after exposing a Pd(100) surface at 170 K to various amounts of H_2. Exposures (corresponding to increasing desorption trace areas): 0, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 2.0, 2.4, 3.6, 4.4, 6 L (1 L $\triangleq 10^{-6}$ Torr s). (The "0 L" curve represents the amount of hydrogen adsorbed from the residual gas atmosphere during the cooling down period.)

coverage, so that the data of fig. 1 may be converted into the variation of the *relative* exposure with coverage. Calibration of the coverage scale was achieved through the LEED data to be described below: Maximum intensity of the diffraction spots of the c(2 X 2) structure was associated with $\theta = 0.5$. (The coverage θ is defined as the ratio of the number of adsorbed H atoms over the number of Pd atoms in the topmost layer.) A quite similar result (however, associated with a rather high degree of uncertainty) was derived from a quantitative analysis of the thermal desorption spectra by taking into account the effective pumping speed and the surface area. As a consequence the low-temperature shoulder in fig. 1 is growing up at coverages above $\theta = 1$ and the saturation coverage results to be about $\theta_{max} = 1.35$, corresponding to a maximum density of adsorbed hydrogen atoms of about $n_s = 1.8 \times 10^{15}$ cm^{-2} at 170 K. Lowering the adsorption temperature to 100 K enables an even slightly higher maximum coverage.

This result (namely a saturation density beyond $\theta = 1$) appears at a first sight somewhat surprising, but is by no means unique: With the system H/W(100) for example $\theta_{max} = 2$ and $n_{s,max} = 2 \times 10^{15}$ cm^{-2} were found [12,13]. The saturation coverage on Ni(111) is near $\theta = 1$ which corresponds to a similar density of about 1.9×10^{15} cm^{-2} [16].

The maximum temperature, T_{max}, of the main TDS peak is at first shifting towards lower values with increasing coverage – indicating second-order desorption

kinetics – but then remains practically constant at $T_{max} = 360$ K. This latter effect suggests pseudo-first-order desorption with an activation energy independent of coverage. The reason for this effect will be discussed later. If a "normal" preexponential for first-order desorption of 10^{13} s^{-1} (corresponding to about 10^{-2} particles cm^2 s^{-1} for second-order desorption) is assumed, from T_{max} the activation energy for desorption is roughly estimated to be about 22 kcal/mole. The shoulder emerging above $\theta = 1$ indicates that this energy is obviously decreasing at high coverages.

The variation of the work function with exposure is reproduced in fig. 2. $\Delta\phi$ increases continuously up to a saturation value of $\Delta\phi_{max} = 0.20$ eV. By using the coverage versus exposure relation as derived from fig. 1 (including the absolute coverage calibration), these data may be converted into fig. 3, exhibiting the variation of the work function with coverage. According to this figure $\Delta\phi$ rises proportional to coverage up to $\theta \approx 0.9$, and from the relation $\mu_s = \Delta\phi\epsilon_0/n_s$, a (constant) dipole moment for the adsorbate complex $\mu_s = 2.1 \times 10^{-2}$ Debye is derived in this range. Above $\theta \approx 0.9$ the slope increases, indicating also an increase of the *average* dipole moment. This is just the range where the TDS data exhibit the emergence of the low-temperature shoulder, thereby suggesting that this state is associated with a higher dipole moment. At saturation an average dipole moment $\mu = \Delta\phi_{max}\epsilon_0/n_{s,max} = 3.1 \times 10^{-2}$ Debye results, whereby it is an open question whether all adsorbed particles have identical dipole moments or if the surface is composed of two types of adsorbed species (e.g. occupying two different kinds of adsorption sites).

Differentiation of the coverage versus exposure-relation yields the sticking coefficient s as a function of coverage. The initial sticking coefficient results to be rather high, $s_0 \approx 0.5$. This number is on the other hand relatively inaccurate due to the uncertainties of the pressure gauge calibration underlying the determination of the exposure. The relative sticking coefficient, s/s_0, is however not affected by this

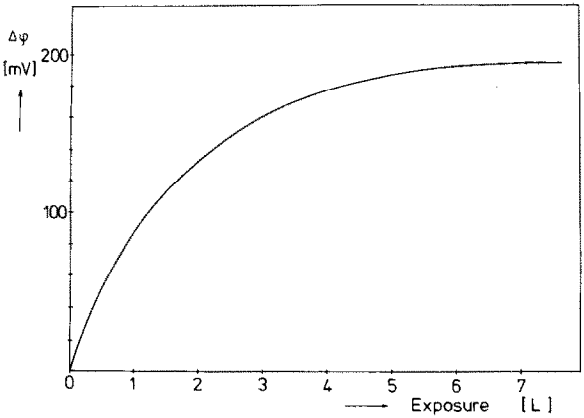


Fig. 2. Variation of the work function change, $\Delta\phi$, with H_2 exposure at 170 K.

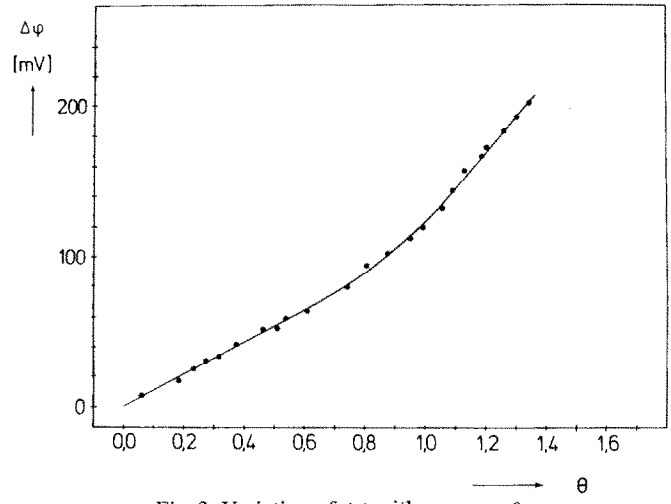


Fig. 3. Variation of $\Delta\phi$ with coverage θ.

problem and the resulting variation of s/s_0 with θ is reproduced in fig. 4.

Since the work function change was found to be a unique function of coverage this quantity may also be used to determine equilibrium data as in our previous work [2]. Fig. 5 shows a series of adsorption isobars, $\Delta\phi$ versus temperature at constant pressures. (It should be mentioned that all these data were obtained at temperatures far above the order–disorder transitions. Eventual variations of $\Delta\phi$ with T at constant θ – as observed with CO/Pd(100) near $\theta = 0.5$ [11] – are therefore negligible).

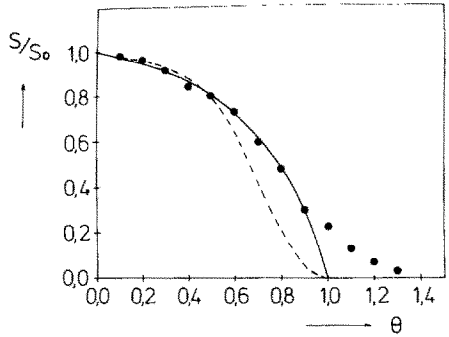


Fig. 4. Variation of the relative sticking coefficient, s/s_0, with coverage at 170 K. Filled circles: experimental data. Broken line: second-order precursor model. Full line: first-order precursor model.

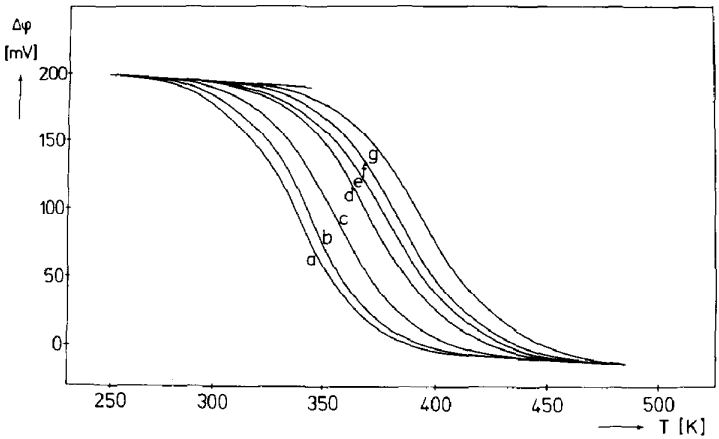


Fig. 5. Adsorption isobars: variation of $\Delta\phi$ with temperature at various H_2 pressures (in Torr): (a) 3×10^{-9}, (b) 5.3×10^{-9}, (c) 1.8×10^{-8}, (d) 8.5×10^{-8}, (e) 1.4×10^{-7}, (f) 2.5×10^{-7}, (g) 5×10^{-7}.

By converting $\Delta\phi$ into θ (fig. 3) and applying the Clausius–Clapeyron equation

$$\left(\frac{\mathrm{d} \ln p}{\mathrm{d}(1/T)}\right)_{\theta=const} = -\frac{E_{ad}}{R}, \tag{1}$$

the isosteric energy of adsorption, E_{ad}, as a function of coverage can be derived. The result is shown in fig. 6. Thereafter $E_{ad} = 24.5$ kcal/mole is practically constant up to $\theta \approx 1$ and then decreases continuously which is in complete agreement with the qualitative conclusions from the thermal desorption measurements.

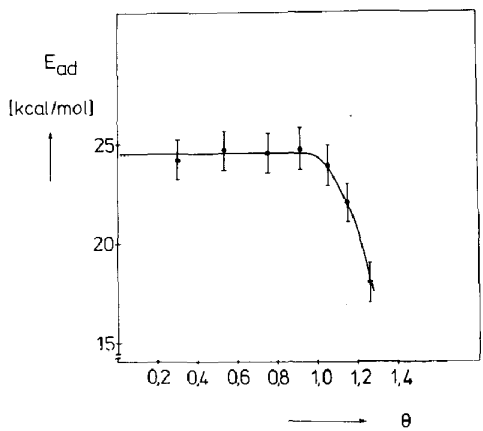


Fig. 6. Isosteric heat of adsorption, E_{ad}, as a function of coverage.

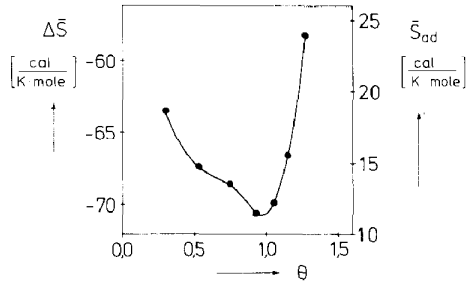


Fig. 7. Differential entropy of adsorption, $\Delta\bar{S}$ (left hand scale), and partial molar entropy of the adsorbed phase, $\bar{S}_{ad}$ (right hand scale), as a function of coverage.

The equilibrium data further yield the differential entropy of adsorption, $\Delta\bar{S}$, through the relation [14]

$$\Delta\bar{S} = \bar{S}_{ad} - S_g^0 = -R \ln\left(\frac{p}{p^0}\right) - \frac{E_{ad}}{R} \tag{2}$$

where $\bar{S}_{ad}$ is the partial molar entropy of the adsorbed phase, and S_g^0 the molar entropy of gaseous H_2 at a standard pressure p^0 which was arbitrarily chosen to be 10^{-8} Torr. The resulting data are reproduced in fig. 7.

In addition, the preexponential factor ν_d of the rate constant for desorption, $k_d = \nu_d \exp(-E_d/RT)$, can be determined. The activation energies for adsorption and desorption, E_a and E_d, are related with the isosteric adsorption energy, E_{ad}, through

$$E_{ad} = E_d - E_a + RT/2 \,. \tag{3}$$

In view of the rather high sticking coefficient at low temperature E_a can safely assumed to be zero, although the temperature dependence of the adsorption kinetics was not studied explicitly, and thus $E_d \approx E_{ad}$. At equilibrium the rates of adsorption and desorption are equal and may be formulated as follows:

$$r_a = sp/(2\pi mkT)^{1/2} \,, \tag{4}$$

$$r_d = k_d n_s^2 = \nu_d \exp(-E_{ad}/RT)\, n_s^2 \,. \tag{5}$$

The latter equation is only valid at low coverages as long as the kinetics of desorption can be described by a second-order rate process (see TDS data). From $r_a = r_d$ follows

$$\nu_d = s \frac{p}{n_s^2 (2\pi mkT)^{1/2}} \exp(E_{ad}/RT) \,, \tag{6}$$

yielding an "effective" preexponential $\nu_d = 8 \times 10^{-3}$ cm^2 (H atom)$^{-1}$ s^{-1}. A more detailed discussion will be presented in section 4.

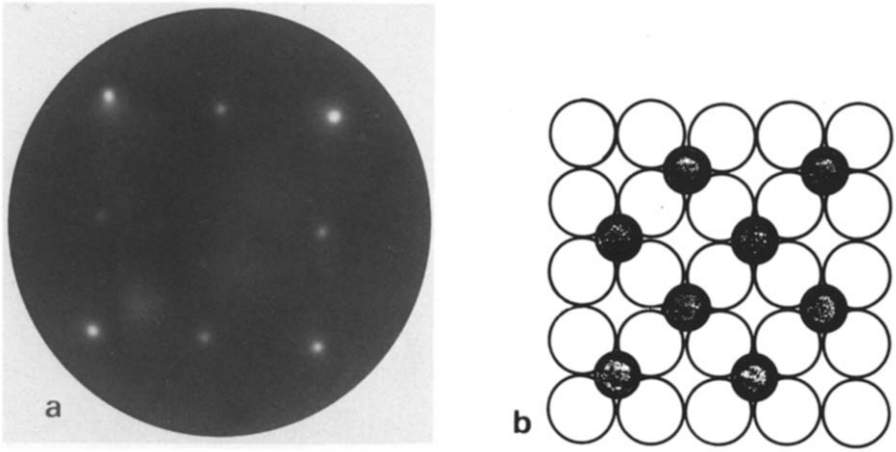


Fig. 8. (a) LEED pattern from the c(2 × 2) structure formed by hydrogen adsorption at low temperatures (b) Structure model for the perfectly ordered c(2 × 2) phase at $\theta = 0.5$.

The LEED pattern exhibits at low temperatures and at intermediate coverages the formation of a c(2 × 2) pattern as reproduced in fig. 8a. The maximum intensity of the "extra" spots is only about 10% of that of the substrate spots which is slightly more than that found for the system H/Ni(111) ($\lesssim$2%) but corresponds to estimates based on dynamical LEED calculations for a non-reconstructed substrate surface [16]. The absence of any noticeable variation of the intensities of the substrate diffraction spots is further support for the conclusion that the LEED pattern is primarily caused by scattering of the H atoms and not by adsorption-induced lateral periodic displacements of the surface atoms, as is e.g. the case with the systems H/Ni(110) [67] or H/W(100) [68]. *I/V* measurements of the (00) beam at room temperature and under 10^{-7} Torr H_2 revealed small shifts of the primary Bragg peaks towards lower energies in a similar manner as observed previously for H/Pd(111) [69] and which is interpreted as being caused by a slight vertical expansion of the topmost atomic layers.

A plausible structure model for the perfectly ordered c(2 × 2) phase is reproduced in fig. 8b. Arguments that justify the assumption of fourfold adsorption sites will be presented in the discussion section.

The variation of the intensity of the $(\frac{1}{2}\frac{1}{2})$ "extra" spot with coverage at 105 K is reproduced in fig. 9. Thereafter this quantity increases nearly linear up to a maximum which is associated with the highest degree of order of the c(2 × 2) structure and thus should correspond to $\theta = 0.5$. This point was indeed used for calibration of the absolute coverage scale as outlined above. Beyond $\theta = 0.5$ the intensity decreases approximately quadratic with coverage and is no longer measurable at $\theta = 1$. No other LEED features were observed in the range of higher coverages. Increasing the temperature at constant coverage causes the additional LEED spots

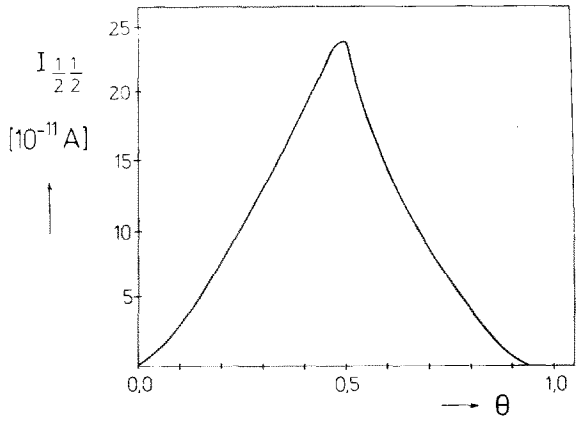


Fig. 9. Variation of the intensity of the half-order LEED spots with coverage at 105 K (U = 52 eV).

to disappear, indicating the occurrence of order-disorder transitions as for example also found with the H/Ni(111) system [15,16]. Fig. 10 shows plots of $I_{1/2\ 1/2}$ as a function of temperature at varying coverages which were corrected by contributions from the Debye–Waller factor. The latter was adjusted in such way that

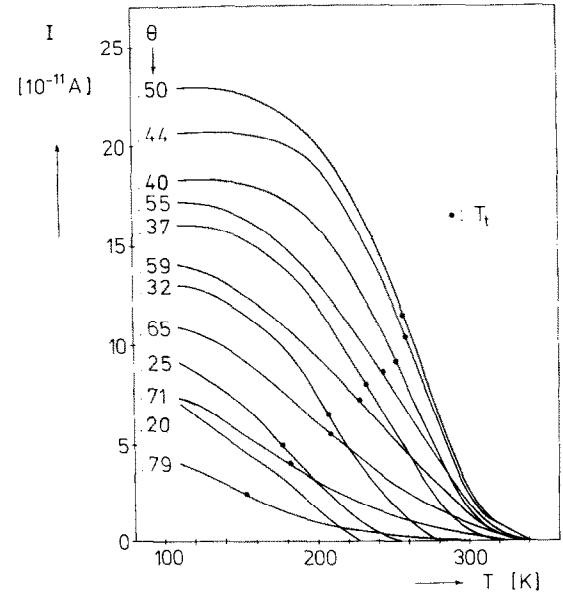


Fig. 10. Variation of the intensity of the half-order spot with temperature (corrected for the "effective" Debye–Waller factor) at various coverages (U = 52 eV). The points of inflection (= order–disorder transition temperatures) are marked.

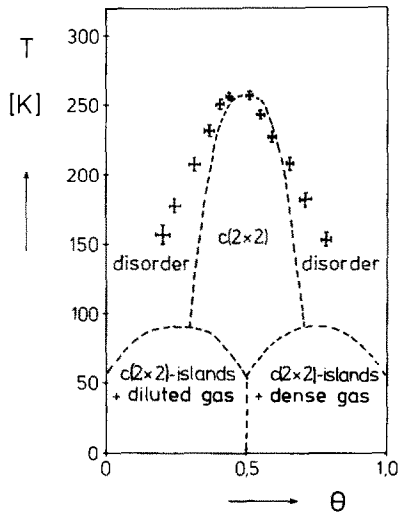


Fig. 11. Experimental phase diagram (data points) and theoretical phase diagram for a square lattice gas model with $w_1 = -2w_2 = 0.51$ kcal/mole [29].

around $\theta = 0.5$ the intensity became independent of temperature in the low temperature range. If the point of inflection of these curves is, in analogy to previous work [15], defined as the corresponding transition temperature T_t, a phase diagram is obtained which is reproduced in fig. 11. This diagram exhibits the region in which long-range order within the adsorbed phase is existent.

4. Discussion

4.1. Comparison with previous work

Due to the easy solubility of hydrogen in palladium this system is in general rather complex. There is, however, general agreement that exposure of a clean Pd surface will at first lead to the build-up of the chemisorbed phase since the adsorption energy is considerably larger than the heat of dissolution (4.6 kcal/mole [1]). In measurements with evaporated Pd films at 78 K, Duš [7] observed with increasing exposure a continuous increase of the work function (β_1^- state) followed by a work function decrease (β_2^+ state) and a rather weakly held (probably molecular) α state. The α state is presumably the precursor for bulk solution [3,4,17,18] and might also be the "precursor" state for dissociative chemisorption. The present study was confined to exposures and pressure/temperature conditions where only a work function increase occurs, i.e where dissolution can still be neglected. The adsorption energy for Pd(100) (24.5 kcal/mole) is in rather good agreement with

the initial adsorption energies reported for polycrystalline Pd surfaces: Beeck [19] determined by calorimetry 26 kcal/mole, from thermal desorption data Aldag and Schmidt [3] derived a value of 25 kcal/mole, and Eley and Pearson [4] of 24 kcal/mole. By means of electrochemical methods Vert et al. [20] derived a value of about 27 kcal/mole. The present data also compare well with the isosteric heats of adsorption derived for Pd(111) (20.8 kcal/mole) and Pd(110) (24.4 kcal/mole) [2] and demonstrate that the strength of the metal–hydrogen bond is only very little affected by the crystallographic orientation of the surface as for example already reported for Ni [21] or W [22,23]. The maximum increase of the work function for Pd(100) (0.20 eV) lies between the values for the more open (110) plane (0.36 eV) and that for the most densely packed (111) surface (0.18 eV) [2].

4.2. Surface geometry and adsorbate interactions

The structure model of fig. 8b proposes that with the c(2 × 2) structure the hydrogen atoms are located in fourfold sites. This view appears to be rather plausible for the following reasons:

(i) This configuration corresponds to the octahedral sites which are occupied by H atoms dissolved in the bulk of fcc Pd [24,25]. Recent neutron diffraction studies revealed at low temperatures the existence of an ordered bulk phase where the hydrogen atoms are distributed in planes corresponding to the c(2 × 2) surface phase [70].

(ii) Atomic adsorption on low-index metal surfaces takes usually place on sites of high symmetry and coordination [26]. With the system H/Ni(100) Andersson [49] concluded on the occupation of fourfold sites from data on the vibrational frequency. Confirmation of this model has of course to wait for a complete LEED intensity analysis. (That this problem can be solved for a hydrogen overlayer has recently been demonstrated [16].)

The adsorbate dipole moment has been found to be constant nearly up to $\theta = 1$, irrespective of the degree of order. This indicates that over this whole range of coverages the adsorbed hydrogen atoms occupy sites with identical local (presumably four-fold) symmetry and that the adsorbate dipole moment is not affected by the degree of occupation of neighboring sites. Since at $\theta = 1$ all four-fold sites will be occupied, beyond this coverage either two types of sites (e.g. four-fold and on-top) or bridge sites have to be occupied. (The concentration of the bridge sites is twice as large as that of the surface atoms, so that these will be saturated at $\theta = 2$. This is just the saturation coverage observed with the system H/W(100) where under these conditions indeed only adsorption on twofold bridge sites was concluded on the basis of vibrational spectroscopic measurements [13,27].) In both cases the average adsorbate dipole moment should be different from the low-coverage value, which is in agreement with the present observations. The question whether for $\theta > 1$, one or two types of site are occupied can probably be decided by investigating the vibration spectra. The decrease of the adsorption energy above

$\theta = 1$ (fig. 6), which is also suggested by the low-temperature shoulder in the thermal desorption spectra, is presumably caused by two effects, the occupation of energetically less favorable adsorption sites and the continuous influence of repulsive interaction between neighboring adsorbates.

The occurrence of a c(2 × 2) structure suggests at lower coverages the operation of repulsive interactions between nearest neighbors on identical sites and from the qualitative shape of the phase diagram one could conclude that the present system might be an ideal candidate for the two-dimensional Ising model with just nearest-neighbor repulsions. Statistical treatments of lattice-gas systems on square lattices with pairwise repulsions between nearest neighbors have been described in the literature [28,29]. It can, however, easily be seen that this picture is too simple: For systems with only nearest-neighbor repulsions, w_1, long-range order should completely break down even at $T = 0$ K for $\theta \leqslant 0.35$ and $\theta \geqslant 0.65$. That this is indeed not the case can be seen from fig. 9. These data were recorded at 105 K. By using the plots of fig. 10 the intensities can approximately be extrapolated to temperatures below 100 K and in this case $I_{1/2\ 1/2}$ increases practically linearly with coverage up to $\theta = \frac{1}{2}$. This indicates island formation [48] for which attractive interactions between next-nearest neighbors, w_2, are mandatory.

The interactions between hydrogen atoms adsorbed on a transition metal surface are certainly dominated by indirect (i.e. "through bond") effects [30,31]. In a model calculation for a cubic surface Einstein and Schrieffer [31] have shown that under reasonable conditions the nearest-neighbor interaction will be repulsive, while the next-nearest is attractive. This conclusion has been confirmed in later calculations [32] and is obviously characteristic for the present system as will be shown below.

An estimate on the order of magnitude of the interaction energies can be made on the basis of solutions for the two-dimensional Ising model. From the phase diagram (fig. 11) the critical temperature is obtained to be $T_c = 260$ K. If only repulsions w_1 between nearest neighbors would exist (which is not the case), T_c is related with w_1 through $w_1 = 1.76kT_c$ [33], i.e. $w_1 = 0.91$ kcal/mole. For non-vanishing next-nearest neighbor attractions w_2 (which corresponds to the anti-ferromagnetic case) the interaction parameters are approximately related with T_c through [34]

$$\exp(w_1/2kT_c) = 2\exp(-w_2/2kT_c) + \exp(-w_2/kT_c)\ ; \qquad (7)$$

w_2 stabilizes the long-range order [17]. Assuming $|w_2| = \frac{1}{8}w_1$, we get $w_1 \approx 1.5kT_c = 0.77$ kcal/mole; for $|w_2| = \frac{1}{2}w_1$, $w_1 \approx 1.0kT_c = 0.51$ kcal/mole; and for $|w_2| = w_1$, $w_1 \approx 0.75kT_c = 0.39$ kcal/mole. The numerical result for $|w_2| = \frac{1}{2}w_1$ is also in agreement with Monte Carlo calculations [29]. For a first approximation it is thus concluded that the repulsive interaction energy between nearest neighbors is about $w_1 = 0.6 \pm 0.2$ kcal/mole (the upper limit holds for weak nnn attraction, whereas the lower is valid for $w_1 = -w_2$). The attraction between second neighbors (NN) is probably in the range between $w_2 = -0.1$ and -0.4 kcal/mole.

If only nearest-neighbor repulsions of 0.5 kcal/mole would exist the differential adsorption energy would be predicted to decrease continuously by 4 kcal/mole H_2 by increasing the coverage from 0 to 1 if a statistical model as developed by Wang [35] is applied. Fig. 5 shows that this is not the case but that instead E_{ad} remains practically constant within the limts of error. The observation of a c(2 × 2) structure requires, on the other hand, the existence of repulsion between nearest neighbors whose influence on the $E_{ad}(\theta)$ relation is, however, presumably partly compensated by nnn attractions so that the net effect is too small to be observed experimentally.

Binder and Landau [29] determined the complete phase diagram for $w_2 = -\frac{1}{2}w_1$ and the result is reproduced in fig. 11 by a dotted line together with the experimental data. The theoretical range of existence for the c(2 × 2) structure is somewhat narrower than found by experiment. The remaining difference could perhaps partly be due to finite-size effects (which flatten the $I(T)$ curves). However, in the ordered state the half-order spots were rather sharp, i.e. the mean domain diameter is at least of the order of the coherence width of the electron beam (≈50–100 Å). It appears to be more plausible that $|w_2| > \frac{1}{2}w_1$ which effect would increase the stability range of the ordered phase.

In this connection another problem comes into play: Although there is some ambiguity in the determination of the transition temperatures from the $I_{1/2\ 1/2}(T)$ curves, the experimental phase diagram is without doubt not completely symmetric with respect to $\theta = 0.5$. Above $\theta = 0.5$ the order is more rapidly destroyed than below this coverage which is also manifested by the asymmetric shape of fig. 9 showing $I_{1/2\ 1/2}$ as a function of θ at 105 K. Since for a lattice gas on a square substrate with *pairwise* interactions always a phase diagram is predicted which is perfectly symmetric around $\theta = 0.5$ [29,36] such a model is obviously not completely appropriate to describe the present situation. Two possible reasons may be discussed for this deviation:

(i) Above $\theta = 1/2$ the adsorbed phase is no longer a lattice gas with occupation of identical adsorption sites, but new binding states with different local symmetry come into play. That this indeed is the case is demonstrated by the fact that the adlayer is not saturated at $\theta = 1$. On the other hand, the E_{ad}, $\Delta\phi$ and TDS data indicate that appreciable occupation of different sites occurs only above $\theta \approx 0.9$.

(ii) The interactions between adsorbed hydrogen atoms are not strictly pairwise. The phase diagram for the system O/W(110) has been found to be highly asymmetric [37] and was qualitatively explained by assuming the existence of three-particle interactions apart from the normal pairwise interactions [38]. The occurrence of non-pairwise indirect interactions in model systems had already been discussed earlier [39,40]. Extended calculations have recently been performed by Einstein [41] for three-particle ("trio") interactions who showed that their strength might become comparable to those of the second-shortest-range pair interactions. For the present system the lowest order interactions are illustrated by fig. 12. The existence of "trio" interactions means, that in case c the total interaction energy w

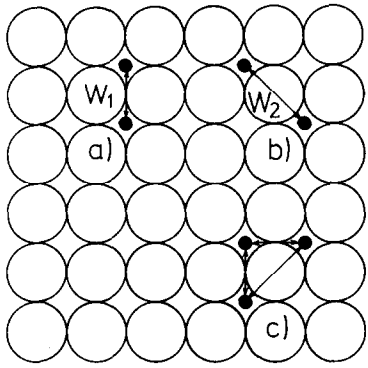


Fig. 12. Types of interactions between adsorbed hydrogen atoms: (a) pairwise repulsion ($w_1 > 0$) between nearest neighbors; (b) pairwise attraction ($w_2 < 0$) between next-nearest neighbors; (c) non-pairwise ("trio") interaction (w).

is obviously not equal to the sum of the pairwise interactions: $w \neq 2w_1 + w_2$. Configurations of this type will be present in higher concentrations at $\theta > 0.5$ than at $\theta < 0.5$ and are therefore presumably responsible for the asymmetry of the phase diagram.

The present data suggest that the effective repulsive interaction is slightly stronger at $\theta > 0.5$ which means that $w > 2w_1 + w_2$. This would cause more rapid destruction of the long-range order at $\theta > 0.5$ which is indeed the case. In concluding this discussion, it is nevertheless believed that the values for the pairwise interactions presented above are at least qualitatively of the correct order of magnitude and that the best agreement between theory and experiment would be obtained for $w_1 = 0.5 \pm 0.1$ kcal/mole, $w_2 = -0.3 \pm 0.1$ kcal/mole.

It is interesting to mention that the existence of attractive interactions between hydrogen atoms dissolved in bulk Pd of about 0.6 kcal/mole is also concluded from the thermodynamic properties of palladium hydrides [1]. The current explanation of this effect is, however, based on an elastic strain model rather than on electronic indirect effects. There is no indication (and also no need) for such an assumption in the case of the chemisorbed phase. In addition, the initial enthalpy of dissolution of hydrogen in Pd (4.6 kcal/mol [42]) is much smaller than the heat of chemisorption (24 kcal/mol), so that no direct connection between the energetics of surface and bulk processes is seen.

4.3. Entropy of the adsorbed phase

The data on the differential entropy of adsorption, $\Delta\bar{S}$, as derived from eq. (2) and reproduced in fig. 7 may be converted into the partial molar entropy of the adsorbed phase $\bar{S}_{ad}$, if the known standard entropy of gaseous H_2 [43] is taken into account. This causes just a shift of the scale as indicated on the right hand side of

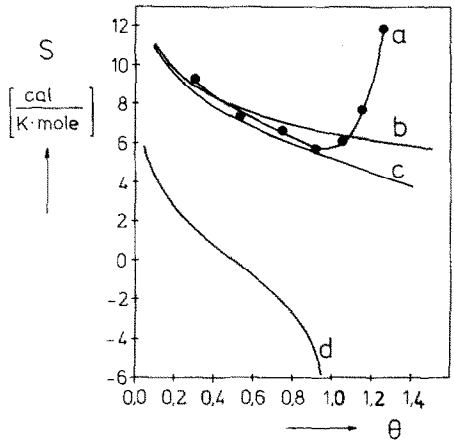


Fig. 13. Differential entropy per mole of adsorbed H atoms as a function of coverage. Curve a: Experimental data. Curve b: Two-dimensional translation of particles each of which covers an area F (= Volmer gas) (eq. (9)). Curve d: Configurational entropy for localized adsorption on a square lattice (eq. (10)).

fig. 7. These data for $\bar{S}_{ad}$ are referred to 1 mole of H_2 and have to be divided by a factor of two in order to obtain the entropy $\bar{S}'_{ad}$ per mole of adsorbed H atoms [44,45]. The resulting data are reproduced in fig. 13, curve a, and have now to be compared with the predictions for the statistical properties of various possible models. The following models, neglecting completely the existence of (weak) lateral interactions between the adatoms, were discussed:

(i) Delocalized adsorption, comprising two-dimensional translation and one vibrational mode of vibration perpendicular to the surface.

(ii) The same as (i) where in addition the effective area occupied by an adsorbed H atom is taken into account to be 2 Å^2 (i.e. $r = 0.8$ Å).

(iii) Localized adsorption on well-defined adsorption sites whose lateral configuration is identical with respect to the substrate lattice. Each H atom then exhibits three vibrational modes (1 perpendicular and 2 parallel to the surface).

The differential entropy for two-dimensional translation of a system of non-interacting particles, $\bar{S}_{tr}$, as a function of coverage was evaluated according to [46]

$$\bar{S}_{tr} = 63.8 + R \ln(MT/N\theta) \text{ [cal/K} \cdot \text{mole]} , \tag{8}$$

and is shown as curve b in fig. 13 (N is the number of adsorption sites). If the area F of an adsorbed H atom is taken into account, eq. (8) transforms into an expression for the so-called Volmer gas [46]

$$\bar{S}_{Vol} = 65.8 + R \ln\left[MT\left(\frac{1}{N\theta} - F\right)\right] - \frac{R}{1 - \theta FN} \tag{9}$$

yielding curve c in fig. 13.

Finally the configurational entropy for localized adsorption (up to $\theta = 1$) neglecting interactions (mean field approximation) was determined according to [46]

$$\bar{S}_{\text{conf}} = -R \ln[\theta/(1-\theta)] \,. \tag{10}$$

The result is reproduced in curve d. The inclusion of interactions would even lower these data.

For all three models the entropy for the vibration normal to the surface has to be added which is related with the frequency ν_i through [47]

$$S_{\text{vib}} = R\left\{\frac{h\nu_i/kT}{\exp(h\nu_i/kT)-1} - \ln[1-\exp(-h\nu_i/kT)]\right\} \,. \tag{11}$$

S_{vib} is in fact the *integral* vibrational entropy which is related with the *differential* entropy through [47]

$$\bar{S}_{\text{vib}} = \theta \, \mathrm{d}S_{\text{vib}}/\mathrm{d}\theta + S_{\text{vib}} \,. \tag{12}$$

As long as ν_i can be assumed to be independent of coverage

$$\mathrm{d}S_{\text{vib}}/\mathrm{d}\theta = 0 \qquad \text{and} \qquad \bar{S}_{\text{vib}} = S_{\text{vib}} \,.$$

Vibrational data for H/Pd(100) have so far not yet been reported. For H/Ni(100), which seems to be similar to the present case, the energy of the perpendicular vibrational mode was found to be $h\nu_1 = 74$ meV [49]. For H/W(100) the energy is $h\nu_1 = 130$ meV [27], for which case the energy of the M–H bond (68 kcal/mole [48]) is similar to that in the present case. With Pd-black samples, a vibrational energy of 125 meV at low hydrogen contents was determined [1,50]. Ratajczykowa [9] studied evaporated Pd films by means of infrared spectroscopy. Only at elevated pressure where already bulk hydrides are formed two bands at 94 and 109 meV were observed which were, however, not attributed to chemisorbed species. From these scarce data it is estimated that the most reliable value is the same as for H/Ni(100), $h\nu_1 \approx 75$ meV, which yields with eq. (11) an entropy contribution $\bar{S}_{\text{vib}}(\nu_1) \approx 0.7$ cal/K · mole at 370 K which is rather small. (This number would only be slightly affected if another value for $h\nu_1$, say 125 meV, would be used.) If this value is added to curve c almost perfect agreement with the experimental data up to $\theta \approx 1$ results.

In order to reconcile the experimental data with the model of localized adsorption the large difference between curves a (experiment) and d (configurational entropy) of about 7 cal/K · mole would have to be almost entirely due to contributions from the two vibrational modes parallel to the surface. The only comparable data are available for H/W(100) where the H atoms are located in twofold sites and give rise to $h\nu_2 = 80$ meV and $h\nu_3 = 160$ meV [27] yielding $\bar{S}_{\text{vib}}(\nu_2 + \nu_3) \approx 0.8$ cal/K · mole. This number is much too small, i.e. both vibrational modes should exhibit much lower frequencies. This would be associated with considerably smaller force constants which in turn requires a rather smooth potential for the motion parallel

to the surface. As a consequence the activation barrier for surface diffusion would be lowered so that even in this picture we would end up with a more or less free two-dimensional translation motion. An energy profile for H/Pd(100) (i.e. the variation of the metal–hydrogen bond strength across the unit cell of the substrate lattice) as evaluated within a semiempirical chemisorption theory [51] indicates indeed that these barriers are rather low. Since on the other hand below 260 K an ordered overlayer structure is observed which can only occur for localized adsorption it is concluded that for $T \lesssim 270$ K the adsorbed layer can be described by a localized model whereas for $T \gtrsim 370$ K (to which temperature the entropy data are corresponding) the adsorbed phase is delocalized and performs two-dimensional translation. This conclusion agrees with previous findings for hydrogen adsorption on various polycrystalline metal surfaces [45,52,53].

Above $\theta \approx 1$ the differential adsorption entropy increases strongly and can obviously no longer be adequately described within the model so far discussed. This effect coincides with the occurrence of the second adsorption state (exhibiting, e.g., a higher dipole moment and a lower adsorption energy). A qualitatively similar effect had been observed previously with the adsorption of hydrogen on polycrystalline Pt films [45]. In this latter case an explanation was proposed in terms of a localized adsorption model in which at $\theta > 1$ additional adsorption sites may be occupied giving rise to an increase of the configurational entropy. This picture is believed to be inadequate for the present situation where even at lower coverages the adsorption has been shown to be already delocalized. It is instead concluded that due to the continuous decrease of the adsorption energy above $\theta = 1$ also the vibrational frequency ν_1 is shifted to lower values which would give rise to an increase of the integral entropy, $S_{\mathrm{vib}}(\nu_1)$, as determined by eq. (11) and to a positive $\mathrm{d}S_{\mathrm{vib}}(\nu_1)/\mathrm{d}\theta$. (This qualitative conclusion would not be affected by the development of coupled vibrations at higher coverages.) Inspection of eq. (12) shows that in this case the term $\theta\,\mathrm{d}S_{\mathrm{vib}}/\mathrm{d}\theta$ might indeed achieve considerable values which is thus made responsible for the pronounced observed increase of the differential entropy at high coverages.

4.4. Kinetics of adsorption and desorption

If compared with other hydrogen/metal systems the initial sticking coefficient for the present case, $s_0 \approx 0.5$, appears to be unusually high. Also the variation of s with coverage as reproduced in fig. 4 is unusual for hydrogen adsorption, but on the other hand quite common, e.g., for CO adsorption [54]. Such a behavior is attributed to the operation of a "precursor" mechanism which was developed initially by Ehrlich [55] and by Kisliuk [56] and has been refined in later studies [57,58] as well as extended also to the kinetics of desorption [59,60].

The present system is described in terms of a precursor state model for dissociative chemisorption which was able to reproduce well the data for N_2/W(100) [57]: (i) H_2 molecules striking the surface are trapped with probability α (= condensation

coefficient) in a molecularly held precursor state, the fraction $1-\alpha$ is reflected back into the gas phase.

(ii) The weakly adsorbed H_2 molecules may either desorb, dissociatively be chemisorbed, or migrate to another site.

(iii) Dissociation is only possible if the H_2 molecule meets two neighboring empty chemisorption sites. Due to the existence of pairwise lateral interactions with energy w ($w>0$ if repulsive) the chemisorbed hydrogen atoms are not randomly distributed over the surface. Within the quasi-chemical approximation [61] the number N_{AA} of nearest neighbor occupied site pairs is given by [35]

$$N_{AA} = \tfrac{1}{4} z N_s \theta \left(1 - \frac{2-2\theta}{\{1-4\theta(1-\theta)[1-\exp(-w/kT)]+1\}^{1/2}}\right), \tag{13}$$

where z is number of possible nearest neighbors (=4 for the present square lattice), N_s the number of surface atoms, and w the interaction energy between nearest neighbors. In the present case not only repulsive interactions between nearest neighbors but also attractions between next-nearest neighbors have been demonstrated to exist. However, both interactions have the same effect, namely to prevent a larger fraction of neighboring adsorption sites to be doubly occupied as compared with a random distribution so that the use of a single "effective" interaction parameter appears to be justified for the present purpose.

From eq. (13) the fraction of the surface exhibiting pairs of neighboring empty sites is [57]

$$f(\theta) = 1 - \theta - \frac{2\theta(1-\theta)}{\{1-4\theta(1-\theta)[1-\exp(-w/kT)]\}^{1/2}+1}\,. \tag{14}$$

The variation of the sticking coefficient with coverage then can be written

$$s/s_0 = [1 + Kf^{-1}(\theta) - 1]^{-1}\,, \tag{15}$$

whereby $K = p'_d/(p_a + p_d)$ (p_a = probability of adsorption over an empty site, p_d and p'_d = probabilities for desorption over empty and occupied sites, respectively) and $s_0 = \alpha/(1 + p_d/p_a)$. By using $w = 0.5$ kcal/mole and $K = 0.05$ the broken line in fig. 4, results which is in fairly good agreement with the experimental data up to about $\theta = 0.6$.

The deviations at higher coverages may be attributed to the following effect: The build-up of a second state above $\theta \approx 0.9$ indicates that the actual attainable maximum coverage is >1. As a consequence $f(\theta)$ as derived from eq. (14) is too small. It is obviously not necessary that a hydrogen molecule meets two empty neighboring lattice sites, but the additional sites occupied by the second state may also enable dissociation. The adsorption kinetics can thereafter approximately be described by pseudo-first-order kinetics with respect to the concentration of empty lattice sites. Using this first-order model (for which $s = s_0(1-\theta)/(1-K\theta)$ [56] and $K = 0.75$, the full line in fig. 4 results which obviously represents a rather good fit to the experimental results up to $\theta = 1$.

For even higher coverages (= second state) the adsorption energy decreases strongly so that the derived sticking coefficients are already affected by the onset of desorption and no further attempt for modeling the data in this range can be made.

It should be mentioned, however, that the basic assumption of the present model, namely that first step the adsorption of molecular hydrogen takes place, is by no means theoretically justified: According to theoretical considerations concerning the energy exchange between the incoming particle and the surface, adsorption of molecular H_2 might be rather improbable [62]. Thus further clarification of the microscopic dynamics is needed before a really reliable model for the kinetics of hydrogen chemisorption can be presented.

With regard to desorption kinetics, instead by using eq. (5), the rate of associative desorption may more adequately be formulated as

$$r_d = d[H_2]/dt = -\tfrac{1}{2}\, dn_s/dt = k'_d N_{AA}\,, \tag{16}$$

if it is assumed that the rate-limiting step consists in the recombination of two neighboring adsorbed H atoms. N_{AA} is approximately described by eq. (13). Without interactions ($w = 0$), eq. (13) simplifies to

$$N_{AA} = n_s\theta = N^{-1} n_s^2\,, \tag{17}$$

whereby N is the number of surface atoms (to which the coverage is referred) and eq. (16) transforms into

$$r_d = k'_d N^{-1} n_s^2 = k_d n_s^2\,, \tag{18}$$

which is equivalent to eq. (5) applied to the analysis of the experimental data. (Note that k'_d has the dimension [s^{-1}] whereas that of k_d is [cm^2 (H atom)$^{-1}$ s^{-1}].)

As soon as interactions ($w > 0$) come into play, N_{AA} will be given by eq. (17) only in the limit of vanishingly small coverages, whereas at high coverages N_{AA} becomes proportional to n_s [63,64], i.e. the desorption kinetics follows first-order kinetics with respect to the coverages. This is exactly the situation reflected by the thermal desorption spectra reproduced in fig. 1.

Since in the present case always $N_{AA} < n_s^2/N$ (i.e. the number of occupied pairs is smaller than in a random distribution), the preexponential for desorption as derived by applying eq. (5) (= eq. (18)) will be different from the "true" exponential resulting from application of eq. (16). With $w = 0.5$ kcal/mole, it turns out that the "true" preexponential at $\theta = 0.1$ should be about twice as large as the number derived on the basis of eq. (5) (8×10^{-3} cm^2 $atom^{-1}$ s^{-1}), i.e. the deviation is not very significant. Its order of magnitude agrees with the data for "normal" hydrogen desorption from other transition metal surfaces (in contrast to H/Pt(111) where an unusually small value was derived) [65].

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